NOOR AHMAD FAIZ KHAN

Computer Science & Engineering Student | AI/ML Researcher | Full-Stack Developer



CONTACT

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ResearchGate Profile

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS, SQL

AI / ML / DL: scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, SHAP, LIME, Pandas, NumPy, Matplotlib

Backend & Web: Spring Boot, Flask, Django, Thymeleaf, REST APIs, Maven

Databases: MySQL, PostgreSQL, MongoDB

Tools & IDEs: Git, GitHub, Google Colab, IntelliJ IDEA, VS Code

Productivity: MS Office (Word, Excel, PowerPoint), Prompt Engineering

SOFT SKILLS

- Teamwork & Leadership
- Effective Communication
- Analytical & Problem Solving
- Adaptability & Time Management

LANGUAGES

- Bengali — Native
- English — Proficient

PROFESSIONAL SUMMARY

Final-year Computer Science & Engineering undergraduate at Southeast University, Dhaka, with hands-on expertise in Data Engineering, Machine and Deep Learning, and full-stack software development across Spring Boot, Flask, and Django. Author of multiple IEEE and Elsevier publications and a contributor to ICML and NeurIPS workshops, with a research focus on interpretable, privacy-aware machine learning for healthcare applications. Experienced in delivering complete technical solutions, from model design and evaluation to deployed web applications, with a strong foundation in research methodology and cross-functional collaboration.

EDUCATION

Bachelor of Science in Computer Science & Engineering (CSE) *July 2023 – Present*
Southeast University, Dhaka, Bangladesh | Final Year

Student ID: 2023100000070 | **CGPA:** 3.82 / 4.00 (Running)

Higher Secondary Certificate (HSC) — Science *2021*
Dhaka City College, Dhaka (English Version) | GPA: 5.00 / 5.00

Secondary School Certificate (SSC) — Science *2019*
BIAM Model School and College, Dhaka (English Version) | GPA: 5.00 / 5.00

RESEARCH PUBLICATIONS

1. **An Ensemble Machine Learning and Explainable AI Framework for Thyroid Disease Classification** — *BECITHCON 2025, IEEE* **[Published]**
2. **Integrating Clinical and Socioeconomic Features in an Explainable AI Model for Paediatric Respiratory Disease Prediction** — *QPAIN 2026, IEEE* **[Published]**
3. **An Explainable Ensemble Machine Learning Framework for Predicting IVIG Resistance in Kawasaki Disease** — *Measurement: Digitalization, Elsevier* **[Published]**
4. **FD-Loss: Supervised Feature Decorrelation as a Scale-Invariant Replacement for Random Dropout** — *ICML 2026 Workshop (MusIML)* **[Published]**
5. **OtosFiNet: An Explainable Deep Learning-Based Approach for Automated Ear Disease Classification Using Otoscopic Images**— *SPICSCON 2026, IEEE* **[Accepted]**
6. **DP-NegGrad+: Differentially Private Machine Unlearning with a Post-Quantization Robustness Check for Diabetes Risk Prediction**— *NeurIPS 2026 Workshop (MusIML)* **[Accepted]**
7. **Rashomon Sets Across Tree Ensembles: Predictive Multiplicity in Gradient Boosting and Random Forests**— *NeurIPS 2026 Workshop (MusIML)* **[Accepted]**
8. **How Much Does Privacy Cost? An Empirical Study of Client-Level Differential Privacy and Quantization in Cross-Silo Federated Clinical Risk Prediction**— *NeurIPS 2026 Workshop (MusIML)* **[Accepted]**
9. **FedRL-FuseNet: An Explainable Federated Reinforcement Learning-Guided Contrast-Enhanced Fusion Network for Brain Tumor MRI Classification**— *Journal of Genetic Engineering and Biotechnology, Elsevier* **[Under Review]**

PROJECTS

ContextEnglish — Contextual English Learning Platform *(Spring Boot, Thymeleaf, PostgreSQL/Supabase)* — Full-stack web app for passive English learning through context-based exercises, with adaptive difficulty and progress tracking.

ShobConnect — Service Marketplace Platform *(Java Spring Boot, Thymeleaf, MongoDB)* — Platform connecting clients with skilled service professionals such as electricians and mechanics, with booking and listing management.

Intelligent Finance Planner *(HTML, CSS, JavaScript)* — Full-stack web app for tracking personal and organizational finances, with an analytics dashboard for income, expense, and savings insights.

Hospital Management System *(JavaFX, MySQL)* — Desktop application for managing patient and staff records with a structured relational database backend.

Library Management System *(Java, Spring Boot)* — Software for managing book inventory, member records, and borrowing workflows.

Apartment Management System *(HTML, CSS, JavaScript)* — Web application for managing apartment and tenant records with an interactive front-end.

Binary-Strike — Browser-Based Fighting Game *(HTML, CSS, JavaScript)* — Two-player canvas-based fighting game with real-time health tracking, round timers, and custom animations.

Lung_CNN_Classification *(Python, TensorFlow/Keras, XGBoost, SHAP, LIME)* — CNN-based lung disease classification pipeline comparing multiple pretrained architectures, with SHAP/LIME explainability for model interpretation.

EXPERIENCE

Research Assistant*2025 – Present*
Southeast University, Dhaka

Private Tutor*May 2022 – Present*
Ahmad Khan Academy, Dhaka

CORE COMPETENCIES

- **Office & Database Tools:** Proficient in MS Word and MS Excel for report drafting, documentation, and structured record-keeping; working knowledge of MS Office suite and Database Management Systems (DBMS) across relational and NoSQL environments.
- **Data Collection & Quality:** Experienced in designing data collection instruments and digital survey/data entry forms, with careful attention to data sampling strategy, data quality assurance, and validation to ensure clean, representative datasets.
- **Data Engineering & Visualization:** Hands-on experience building data pipelines, including ingestion, preprocessing, and feature engineering, with results communicated through clear visualizations using Pandas, Matplotlib, and interactive dashboards.
- **Testing & Observability:** Skilled in testing and reviewing digital tools and platforms, documenting functional feedback, and monitoring data/model pipelines for consistency and reliability.
- **Research Dataset Preparation:** Curated and preprocessed structured datasets for machine learning and LLM-related research, including feature engineering and privacy-aware preprocessing for clinical and tabular data.

ACHIEVEMENTS & INVOLVEMENT

- National Winner — Bangladesh IQ Olympiad 2021 (Feb 2022)
- Bronze Honour — International Astronomy & Astrophysics Competition (IAAC), 2021
- Bronze Honour — International Youth Math Challenge (IYMC), 2021
- Participant — International Collegiate Programming Contest (ICPC)
- Participant — Kaggle Data Science Hackathons
- Sub-Executive — IEEE Southeast University Student Branch & WIE Affinity Group

REFERENCES

Prof. Yusuf Mahbubul Islam, PhD
Vice Chancellor & Professor, Department of CSE, Southeast University, Dhaka

Khandaker Mohammad Mohi Uddin
Assistant Professor, Department of CSE, Southeast University, Dhaka